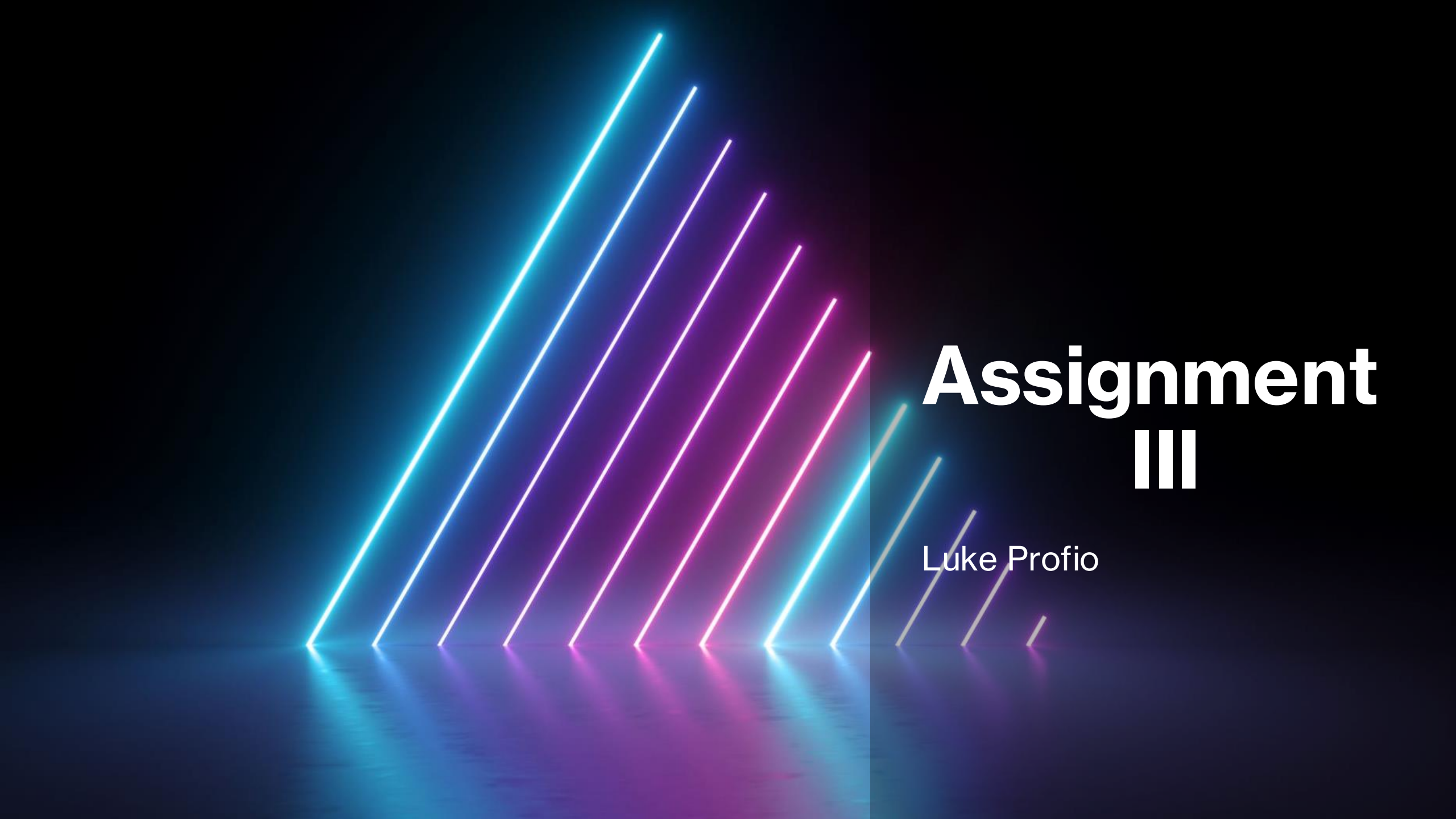


Assignment III

Luke Profio

Extracted Notes

- Here we extract notes from ICD code 4019, patients with Unspecified Essential Hypertension.
- Of these notes, we filter for nutrition by category.
- This filters notes from patients with Unspecified Essential Hypertension for their nutritional panel notes (e.g. glucose levels).

```
ICD_FILTER = [
    '4019'
]
df_filtered = df_diagnoses_icd[df_diagnoses_icd['ICD9_CODE'].isin(ICD_FILTER)]
subj_filtered = df_filtered['ROW_ID'].unique()
print(f"Number of ICD-9 {ICD_FILTER} patients: {subj_filtered.size}")
```

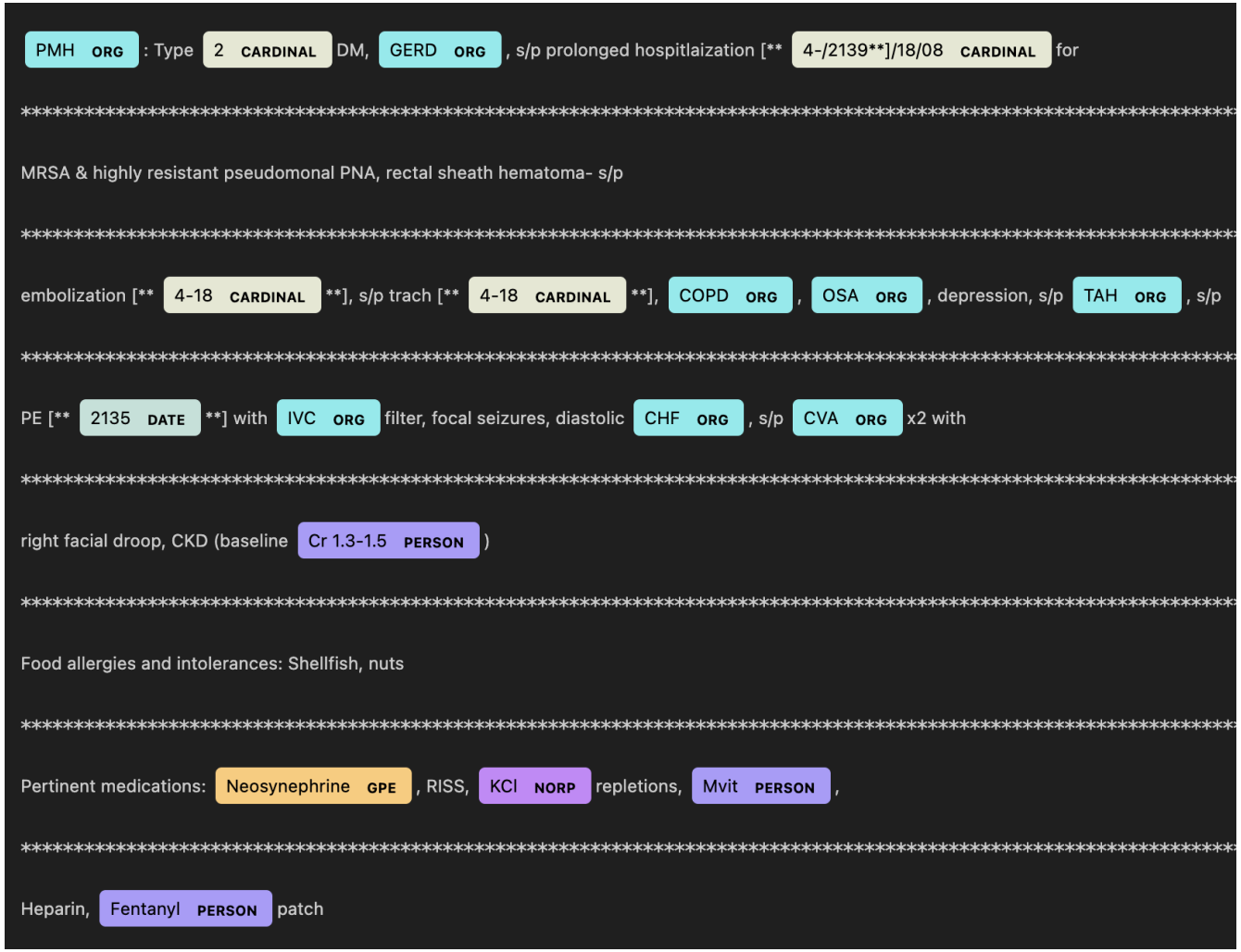
```
Number of ICD-9 ['4019'] patients: 20703
```

```
df_4019_nutrition_notes = df_noteevents_filtered[df_noteevents_filtered['CATEGORY'] == 'Nutrition']
df_4019_nutrition_notes.info()
```

```
Glucose
147 mg/dL
[**2139-7-7**] 01:18 AM
Glucose Finger Stick
193
[**2139-7-7**] 06:00 AM
BUN
54 mg/dL
[**2139-7-7**] 01:18 AM
Creatinine
3.6 mg/dL
[**2139-7-7**] 01:18 AM
Sodium
130 mEq/L
[**2139-7-7**] 01:18 AM
Potassium
3.6 mEq/L
[**2139-7-7**] 01:18 AM
Chloride
100 mEq/L
[**2139-7-7**] 01:18 AM
Albumin
```

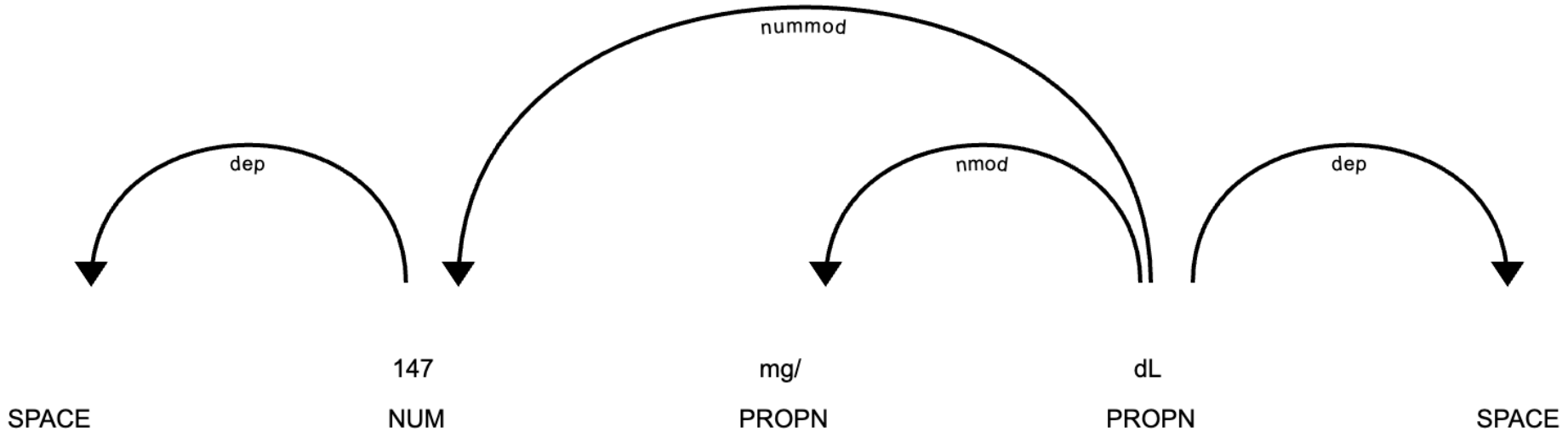
Displacy-Extracted Values (Spacy)

- Here we are provided the generic Spacy-extracted entities.
- Overall, the utility of these entities is low.
- Many entities are inaccurate (e.g. Mvit, or multivitamin, marked as a ‘person’).



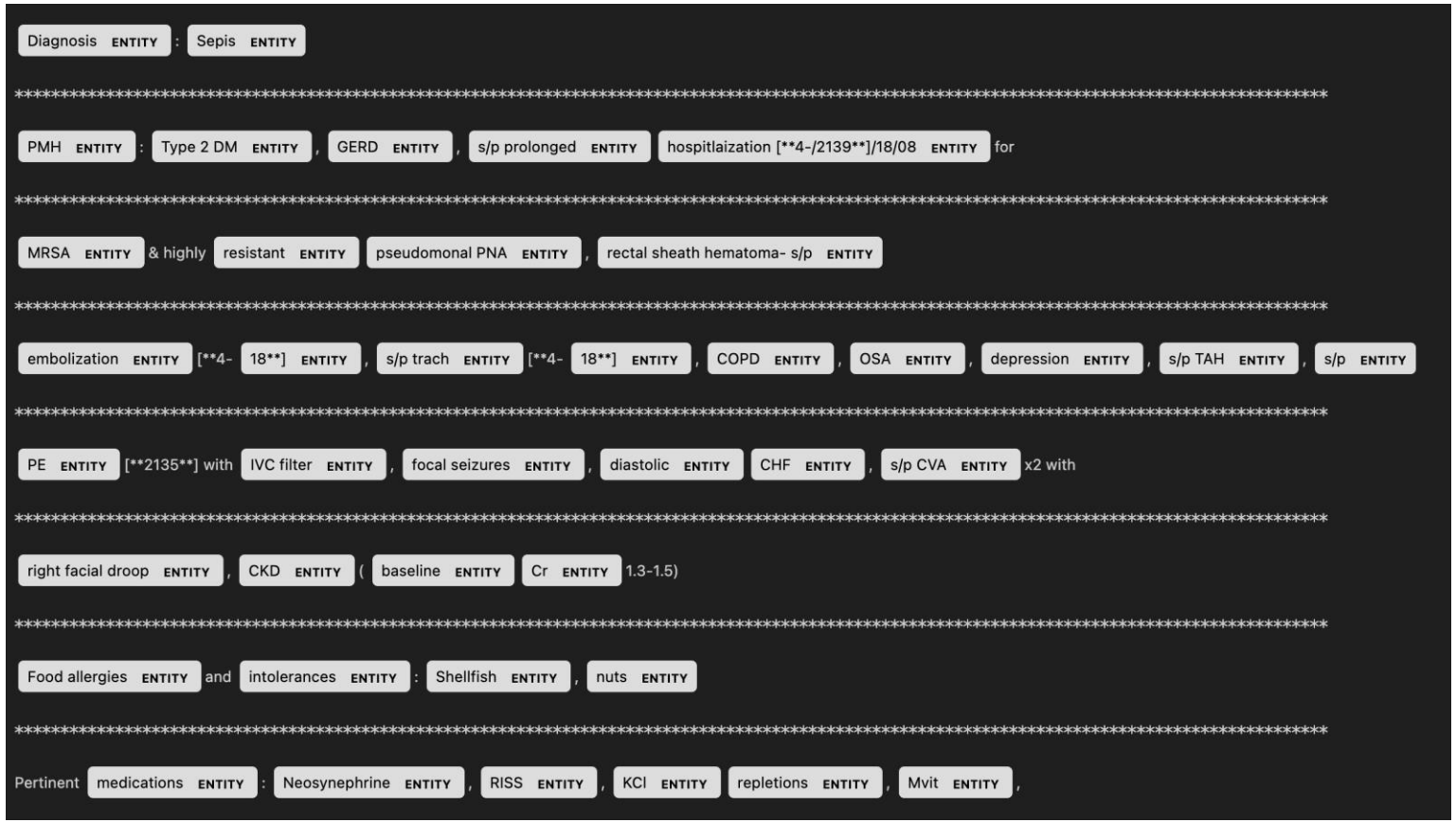
Sentence Dependency Tree (Spacy)

- Here we are provided a dependency tree on the sentence level.
- Spacy effectively models numeric relations relevant to nutritional values, such as metrics.
- For example, 147 mg/dL is modeled as follows:



Displacy-Extracted Values (SciSpacy)

- Prior to more enriched SciSpacy extraction, we mark all relevant entities.
- This process is efficient, with a high percentage capture of appropriate entities (e.g. chemicals).
- Note: my notebook may not have been fully executed due to a memory/runtime constraint.



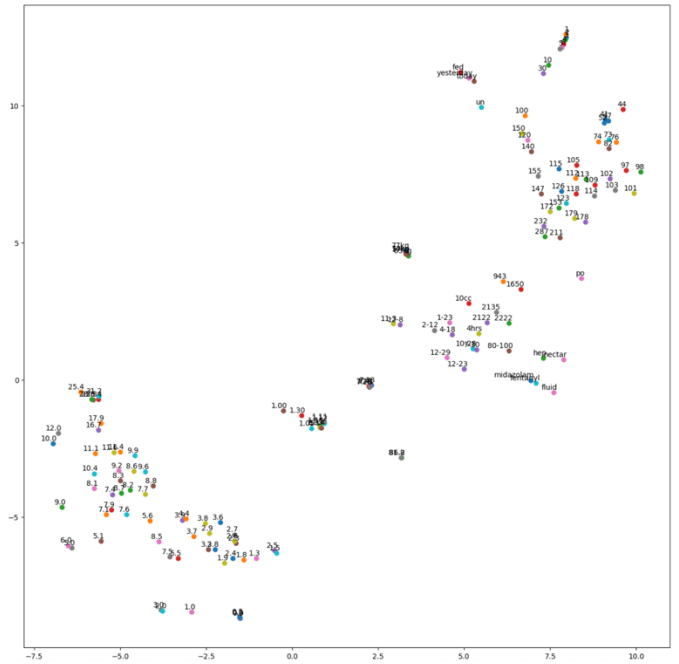
Displacy-Extracted Values (SciSpacy)

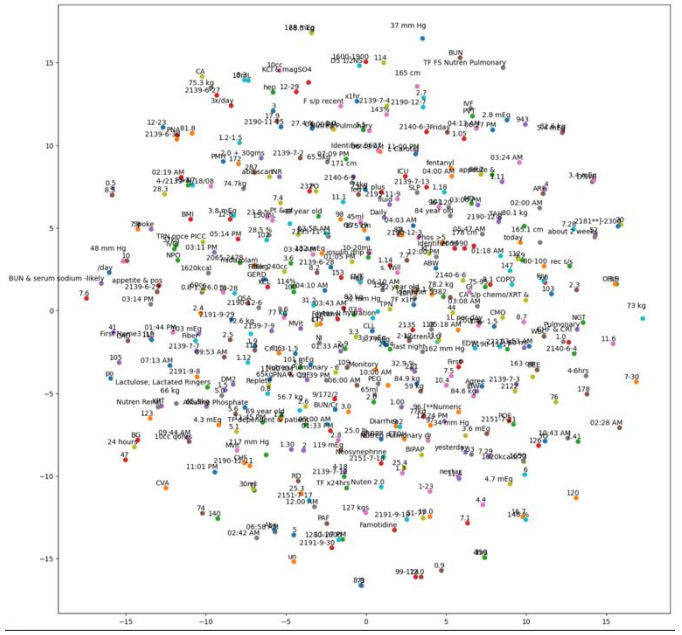
- As seen below, SciSpacy is significantly more effective at extracting relevant values, such as diseases (e.g. COPD) and chemicals (e.g. Magnesium).
- This tagging is critical to monitor nutritional values and infer broader trends (e.g



Word2Vec tSNE Plot (Spacy)

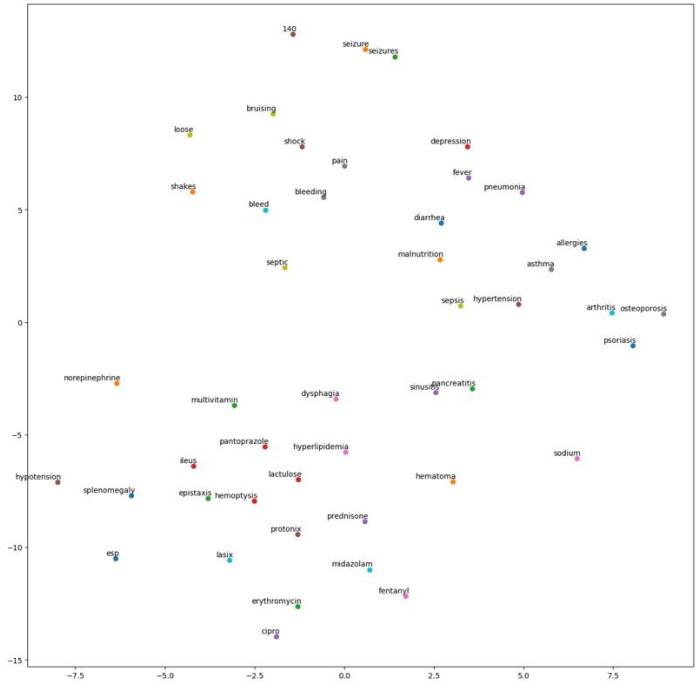
- Given the total amount of numeric outputs and disparate note content, there is not much utility nor grouping in the tSNE plot.
- This could be due to Spacy's limited entity extraction explored earlier.

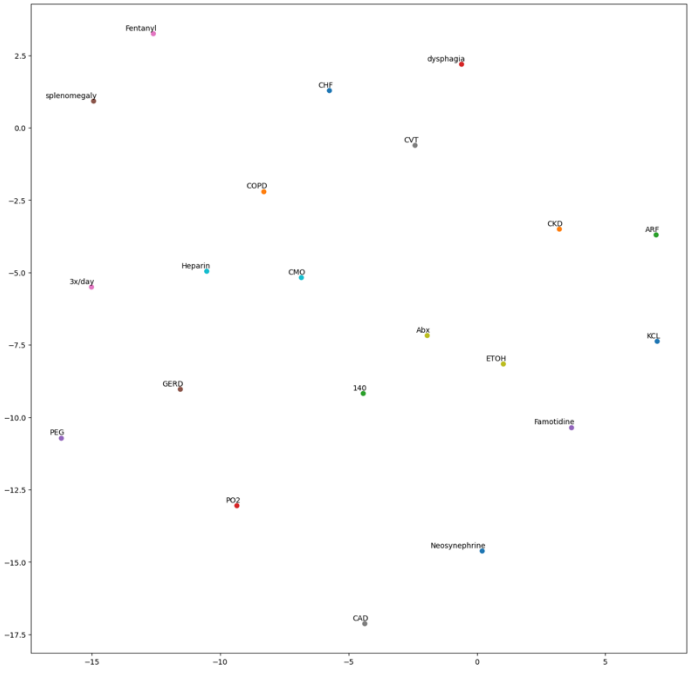




Word2Vec tSNE Plot (SciSpacy)

- There is a clear improvement in the tSNE plots of SciSpacy.
- Entities are no longer scattered, and have a more logical grouping (e.g. fever nearby pneumonia).
- Overall there are significantly less entities, and they are of higher quality (e.g. chemicals, diseases).





ClinicalBert tSNE Plot

- Embeddings Tensor Shape: ([25, 768])
- Notes were read in, tokenized, and passed through ClinicalBERT prior to mean-pool in the last hidden states to get a fixed-size vector per note.
- To visualize with t-SNE, we reduce the stack of embeddings from 768 down to 2 dimensions via PCA, with a scatter plot to reveal any semantic clusters.
- Given clusters are equally apart from one another, the data is relatively homogenous with no clear subpopulations.

